

CKN v1.4.2

Mathematical Audit and Revision Disposition

Companion to the v1.4.2 Release Candidate
September 5, 2026

1. Finding and scope

The revised mathematics proves a substantive but restricted result: a local geometric inequality and a local connection to an independently specified task operation jointly imply persistence of initialized operation authority and preservation of actual task traces. An explicit family of abstract transition systems satisfies those hypotheses with nonzero input coupling. A numerical instance preserves its task while an unrestored comparator changes the initialized operation under the same input history. A second instance identifies the geometry with the actual task state, showing that the hypotheses do not require a separate, behaviorally inert authority coordinate.

This is more than preservation of an arbitrary positive register. It is also less than a theory of general privileged instructions in language models. A fixed reference operation can itself contain input-triggered behavior that a human describes as an override. Preserving that operation reproduces its behavior. The theorem does not identify or repair instruction hierarchy inside an arbitrary reference model. The main paper now makes this limitation explicit.

Disposition: Theorems 6.1 and 6.2, Proposition 7.1, and Proposition 7.4 are valid under their stated hypotheses. Their conjunction has nontrivial mathematical instances, including one with $q = g = \text{id}$. No neural attainability, training method, practical capability comparison, or new general verification algorithm is established. The remaining central research gap is the connection between specified operation authority and useful privileged model conditioning.

This audit checks the v1.4.2 release candidate and preserves the earlier proof checks and report dispositions. Section 7 distinguishes the reviews of v1.3.0, v1.4.0, and v1.4.1. The closing revision retains the core theorems, adds a conditional factorization observation, makes the decoder obligation explicit, and strengthens the optional comparison lemma's domain hypotheses. Source object numbers in the historical register refer to v1.2.0. Mathematical examples below are attempts to falsify implications or their interpretation, not implementation proposals. The review is an inspection of written mathematics, not machine-checked formal verification or a parameter audit.

Correction to the previous audit

The v1.3.0 audit correctly verified its scalar implications but underplayed the interpretive gap. It treated a concrete privilege relation as a realization parameter rather than recognizing that the formal conditions failed to connect that relation adequately to the computation under review. Its final claim that this was merely a realization limitation was too broad. The missing connection was a limitation of the theory's stated meaning. Version 1.4.0 repairs that connection for specified task operations; it does not resolve every broader meaning of privilege.

2. Proof check: geometric persistence

Fix a privileged configuration and suppress its subscript. Let

$$v(\Sigma) = \delta(g(\Sigma), z), \quad R = \{\Sigma : v(\Sigma) < a\}, \quad c = \frac{d}{1-\lambda}, \quad M = \max\{v(\Sigma_0), c\} < a,$$

where $0 \leq \lambda < 1$, $d \geq 0$, and $a > 0$. The transition inequality is required for every state in R and every event:

$$v(F_\Theta(\Sigma, e)) \leq \lambda v(\Sigma) + d.$$

The inequality is local in time, not a statement that only nearby states or typical events are covered. The quantifier covers the entire stated region and event alphabet. This is a stronger obligation than checking observed transitions, but it is an obligation directly on the transition map.

Base case. At $t=0$, the finite sum is empty and the proposed bound equals $v(\Sigma_0) \leq M < a$. Thus the transition inequality applies to the first actual step.

Induction. Suppose the bound holds at step t . Then $\Sigma_t \in R$, so

$$\begin{aligned} v(\Sigma_{t+1}) &\leq \lambda v(\Sigma_t) + d \\ &\leq \lambda^{t+1} v(\Sigma_0) + d \sum_{j=0}^t \lambda^j. \end{aligned}$$

For every nonnegative integer t ,

$$\lambda^t v(\Sigma_0) + d \sum_{j=0}^{t-1} \lambda^j = \lambda^t v(\Sigma_0) + (1 - \lambda^t) c \leq M < a.$$

At $t=0$, the coefficient λ^0 is the empty product 1, including when $\lambda=0$. The right-hand side is a convex combination of two quantities below a . This proves the claimed bound and closes the induction without assuming trajectory containment in advance.

What is established. The run stays in R for every finite step of every event history, with uniform positive margin at least $a - M$. The whole task state need not be bounded or contractive. Different privileged regimes need not approach one another. The residual c is an asymptotic upper bound on deviation, not a lower bound on actual deviation: cancellation or zero disturbance can give smaller values.

Boundary attacks. Initialization cannot be omitted. A transition preserving a region says nothing about a start outside it. With $\lambda = 1$ and positive residual disturbance, repeated increments can accumulate without bound. At $M = a$, the theorem's strict certificate does not apply; this is not a proof of failure. For example, $v_{t+1} = \lambda v_t + (1 - \lambda)a$ with $0 < \lambda < 1$ and $v_0 < a$ remains strictly below a at every finite step while tending to the boundary. The sufficient strict certificate intentionally excludes this case.

Conversely, dropping the budget can permit escape: $v_{t+1} = v_t/2 + 3/4$, $v_0 = 0$, $a = 1$ reaches $9/8$ at step two. No statement about small individual inputs repairs an absent full-transition bound.

3. Proof check: preservation of the actual task

The reference is fixed independently:

$$x_{t+1} = B_p(x_t, e_t), \quad y_{t+1}^p = H_p(x_t, e_t).$$

The candidate must satisfy, throughout R_p and for every event,

$$q(F_\Theta(\Sigma, e)) = B_p(q(\Sigma), e), \quad Y_\Theta(\Sigma, e) = H_p(q(\Sigma), e),$$

with $q(\Sigma_0) = x_0^p$. The first identity connects actual task-state evolution to the reference. The second connects the actual emitted observation to that same task computation. Neither is inferred merely from geometric persistence or builder control of parameters.

Induction. The task states agree initially. Theorem 6.1 ensures the current actual state lies in the region where both identities hold. If $q(\Sigma_t) = x_t$, then

$$q(\Sigma_{t+1}) = B_p(x_t, e_t) = x_{t+1}, \quad y_{t+1} = H_p(x_t, e_t) = y_{t+1}^p.$$

This proves every finite observation trace identical on a common event history. It is an ordinary exact simulation argument. It preserves properties of the declared traces without assigning them an absolute quality or correctness score.

Operation identity here is extensional and relative to the declared domain. Different internal implementations can represent the same operation. The theorem establishes no unique causal implementation and does not distinguish operations that agree on every covered state and event.

Attempts to make this vacuous. A stable register can coexist with arbitrary task behavior, but then task matching fails for an independently specified operation that is being displaced. An unused copy of the reference can satisfy the state identity while the actual output degenerates, but then the readout identity fails. An auxiliary header can be preserved while content changes; that establishes only a header claim unless the observation includes the content. A constant reference can be preserved, but supplies no substantive capability premise.

The two non-degeneration checks in Section 4.3 are necessary safeguards for an informative instance, not a general characterization of competence. Under a fixed decoding policy and exogenous randomness law, some changes to admitted task input must change the actual task-output trace law; some changes to initialized operation must also change that law for a common admitted input history. In a deterministic instance these reduce to differences in the actual traces. Randomness alone cannot witness responsiveness. These checks alone still permit simple tasks. Exact preservation of a substantively chosen reference, over an independently declared domain and observation, carries the stronger capability conclusion.

Congruence, not an inert-geometry theorem. For two states in the same R_p with equal q values and a common event, the operational identities imply equal next q values and equal current task observations. This is a valid consequence of the hypotheses. It does not imply that g lies outside q , that q has a nontrivial fiber, or that geometric state cannot affect task output. If q is injective, equal q simply means equal actual state. Describing a literal skew-product structure requires additional representation assumptions. Proposition 7.4 supplies $q = g = \text{id}$, where the geometry is the observed evolving task state.

Factorization: a valid conditional reduction. Suppose $h: \mathcal{X} \rightarrow \mathcal{Z}$ and $g = h \circ q$ on $\mathcal{S}$. Given H-op, for every $\Sigma \in R_p$ and event e ,

$$v_p(F_\Theta(\Sigma, e)) = \delta(h(B_p(q(\Sigma), e)), z_p).$$

Consequently H-geo is equivalent to

$$\delta(h(B_p(x, e)), z_p) \leq \lambda_p \delta(h(x), z_p) + d_p$$

for all $x \in q(R_p)$ and all e . The equivalence follows by substitution in one direction and choosing a preimage of each $x \in q(R_p)$ in the other. It does not assert that $q(R_p) = \mathcal{X}$. Establishing H-op, the factorization, and the relevant image remains part of the candidate claim. Factorization only on R_p would be insufficient before proving persistence unless it also holds on the one-step image. Changing the candidate's input gain while holding the reference fixed can violate H-op even when the reference still satisfies the displayed bound.

The trajectory consequence is $\delta(h(x_t), z_p) \leq M_p$. This confines the chosen observable; it does not by itself bound task complexity, output diversity, or capability loss relative to a baseline. Neither this special case nor equal- q congruence implies an independent decomposition of arbitrary mixed geometry into task and fiber constraints. Section 4.5 gives concrete checks of both points.

Initialization can select a nontrivial task through its initial represented state and a fixed interpreter. Therefore demanding unequal transition formulas at a common state is not a representation-independent replacement for observed dependence on the initialized operation. The declared configured task and its actual observations are the objects that must differ.

Overlap check. If geometric regions for distinct configurations overlap, both sets of operational identities must hold on their intersection. Where their represented operations disagree, this is impossible. The paper does not assume arbitrary configurations can share arbitrary regions. Its witness uses disjoint regions, and the general theorem applies only when the jointly specified identities are consistent.

4. Joint satisfiability and the numerical attack

4.1. Proposition 7.1

The abstract state is (r, x) , initialized at (p, x_0^p) with $p \in \{-1, +1\}$. The fixed map is

$$r' = \frac{1}{2}r + \frac{1}{2}\sigma(r) + \frac{1}{4}\psi(x, e), \quad k = \sigma(r'), \quad x' = B_k(x, e), \quad y' = H_k(x, e),$$

where $|\psi| \leq 1$ and $\sigma(0) = +1$. For $p = +1$, $R_p = (0, 2) \times \mathcal{X}$; for $p = -1$, $R_p = (-2, 0) \times \mathcal{X}$. Thus $\sigma(r) = p$ throughout the relevant region, giving

$$|r' - p| \leq \frac{1}{2}|r - p| + \frac{1}{4} < \frac{3}{4} < 1.$$

The next authority coordinate has sign p . Consequently the actual task transition and output use B_p, H_p . The local constants are $\lambda = 1/2$, $d = 1/4$, $a = 1$, and $v_0 = 0$, hence $M = 1/2 < 1$. The trajectory estimate is

$$|r_t - p| \leq \frac{1}{2}(1 - 2^{-t}) < \frac{1}{2}.$$

The proof covers both privileged configurations. Neither the sign convention at zero nor discontinuity there causes a problem: the certified trajectory stays a positive distance from zero. The map is total, but no continuity, differentiability, or neural representability is claimed.

Input coupling and initial privilege. Choosing $\psi(x, u) = u$ gives a nonzero coefficient $1/4$ from admitted input to the authority coordinate. The authority coordinate then selects the actual task operation. Both links matter. The same fixed map runs at every step; p occurs only in initialization. Repeated public copies of the opposite root value are included in $u \in [-1, 1]$. There is no absent influence path or recurring external injection of the privileged configuration in this mathematical witness.

4.2. Exact comparator arithmetic

Take $p = +1$ and repeat $u = -1$. In the protected run,

$$r_{t+1} = \frac{1}{2}r_t + \frac{1}{4}, \quad r_0 = 1, \quad r_t = \frac{1}{2} + 2^{-(t+1)}.$$

The unrestored comparator has $r_t^0 = 1 - t/4$. The emitted task is $\sigma(r_{t+1})u_t$ in each system.

Inputs processed	Protected authority coordinate	Comparator authority coordinate
0	1	1
1	3/4	3/4
2	5/8	1/2
3	9/16	1/4
4	17/32	0
5	33/64	-1/4

At step four, the comparator is on the boundary but its declared tie convention still selects $+1$. At step five, it emits $(-1)(-1) = +1$, while the protected system emits $(+1)(-1) = -1$. The latter equals the initialized operation pu . The paper's values and step indexing are correct.

This establishes strict resistance improvement on the specified attack and comparator, with exact task preservation against the fixed-operation reference. Those are two different comparisons. It does not show output identity to the unrestored comparator on every history, nor should it: their divergence on the attack is the demonstrated improvement.

The added capability workload also supports a comparison against that same unrestored system. Write $S_t = \sum u_j$ and fix all histories satisfying $|S_t| \leq 2$ at every prefix. For either root, $r_t^0 = p + S_t/4$ stays within $1/2$ of $p^{j \leq t}$, so its sign is p . This includes the updated state used by each emitted observation, because the condition covers every prefix. Both systems therefore emit pu_t throughout that workload.

The workload includes every single input in $[-1, 1]$ and indefinitely many varying inputs, such as alternating $+1, -1$. It is not an output-selected set. It is nevertheless a restricted capability workload, not a claim about typical language tasks. The repeated opposing input leaves this workload after the third event but remains inside the unchanged resistance domain. Thus one comparator exhibits exact task preservation on a specified nontrivial workload and strict resistance improvement on a declared attack; no security-domain restriction is used to remove that attack.

4.3. Why the arbitrary-transducer lift matters, and where it stops

The proof places no bound on task-state dimension or task complexity and requires no contraction of B_p .

Any total task transducers over the declared common spaces can be substituted into the construction.

Therefore the mathematical consistency result is not limited to multiplication by a sign. It does not require reducing task computation to a constant or a trivial state update.

This is nevertheless a permissive abstract class. Exact branch selection and exact retention of arbitrary reference maps are built into the witness. There is no demonstrated embedding of this construction into a transformer, no cost model, and no evidence of useful language-model weights.

There is also a separate semantic limit. Let a reference operation say, formally, “use operation A until event SWITCH, then use operation B.” The lift preserves that rule, including the switch. A human could call SWITCH an instruction override, but it is not substitution of the formally specified governing rule. This is a counterexample to an attempted extension of the theorem to arbitrary prompt-defined authority, not a counterexample to its actual fixed-operation conclusion. The reference must identify the intended protected operation independently; selecting a baseline because it was initially prompted with p is insufficient.

4.4. Shared task and geometry: $q = g = \text{id}$

Take $\mathcal{S} = \mathcal{X} = \mathcal{Z} = \mathbb{R}$, admitted input $u \in [-1, 1]$, privileged configuration $p \in \{-1, +1\}$, and initial state $r_0 = x_0^p = p$. Declare the reference before the actual map:

$$B_p(x, u) = H_p(x, u) = \frac{1}{2}x + \frac{1}{2}p + \frac{1}{4}u.$$

Use the single fixed actual transition and readout

$$F(r, u) = Y(r, u) = \frac{1}{2}r + \frac{1}{2}\sigma(r) + \frac{1}{4}u, \quad q(r) = g(r) = r,$$

where $\sigma(0) = +1$ as before. The anchor is $z_p = p$, the metric is absolute distance, and $a = 1$. On $R_p = \{r : |r - p| < 1\}$, $\sigma(r) = p$, so both operational identities hold exactly and

$$|F(r, u) - p| \leq \frac{1}{2}|r - p| + \frac{1}{4}.$$

Thus $\lambda = 1/2$, $d = 1/4$, $v_0 = 0$, and $M = 1/2 < 1$. Direct solution gives

$$r_t = p + \frac{1}{4} \sum_{k=0}^{t-1} 2^{-(t-1-k)} u_k, \quad |r_t - p| \leq \frac{1}{2}(1 - 2^{-t}).$$

The emitted value is the actual next task state, $y_{t+1} = r_{t+1}$. At a fixed state in either region, $\partial Y / \partial u = 1/4$ and $\partial Y / \partial r = 1/2$. The output depends on admitted input, retained input history, and initialized operation. There is no unused protected register, independent output header, or nontrivial hidden fiber of q . This explicitly defeats the claim that only trivial solutions are possible when $q = g$, or that every satisfying system must put geometry outside the task representation.

The example is scalar and establishes a limited shared-state instance. Proposition 7.1 separately permits arbitrary task transducers using additional task state. The two examples cannot simply be combined into a theorem of arbitrary task capability with shared geometry. Neither supplies transformer weights, an engineering construction, or a practical capability cost curve.

A deviation bound is not a derivative bound. The hypotheses constrain the resulting deviation, not a derivative with respect to the event. For example, for any $K > 0$, take $F(r, u) = Y(r, u) = \sigma(r) + \frac{1}{4}\sin(Ku)$ with $q = g = \text{id}$ and independently declared $B_p(x, u) = H_p(x, u) = p + \frac{1}{4}\sin(Ku)$. Then $\lambda = 0$, $d = 1/4$, and $a = 1$ satisfy the same conditions, while the input derivative at $u = 0$ is $K/4$. The framework therefore entails neither uniformly small differential input gain nor contraction between arbitrary pairs of task states.

4.5. Proper task quotient and mixed geometry

Both examples below use $p \in \{-1, +1\}$ and admitted input $u \in [-1, 1]$.

A proper shared-state quotient exists. Let $\phi(x, u) = x/2 + \sigma(x)/2 + u/4$, and retain the independently declared reference of Proposition 7.4. On $S = \mathbb{R}^2$, set

$$\begin{aligned} q(r, s) &= g(r, s) = r + s, & F((r, s), u) &= (\phi(r + s, u) - s, s), \\ Y((r, s), u) &= \phi(r + s, u), & \Sigma_0 &= (p, 0). \end{aligned}$$

The map q is surjective and noninjective: $(r + c, s - c)$ has the same image for every c . Direct addition gives $q(F((r, s), u)) = \phi(q(r, s), u)$. On $R_p = \{|r + s - p| < 1\}$, the sign is p , so both operational identities hold and the same bound applies with $\lambda = 1/2$, $d = 1/4$, $a = 1$, and $M = 1/2$. Actual output is the next represented task state. This is a shared-geometry instance with a proper quotient and an independently specified reference; its existence is not an open question. The extra representation freedom adds no task richness and proves no neural realization result.

Mixed dependence need not split into independent bounds. For arbitrary total scalar task references B_p, H_p , set $q(x, w) = x$, $g(x, w) = x + w$, and initialize at $(x_0^p, p - x_0^p)$. Define

$$k = \sigma(x + w), \quad x' = B_k(x, u), \quad w' = k + u/4 - x', \quad Y = H_k(x, u).$$

On $R_p = \{|x + w - p| < 1\}$, $k = p$, so H-op holds and $g(F) = p + u/4$. Thus H-geo holds with $\lambda = 0$, $d = 1/4$, $a = 1$, and $M = 1/4 < 1$. The two coordinate contributions can be arbitrarily large while their sum remains bounded. A decomposition imposing independent bounds on those contributions would add assumptions absent from the theorem. This is a counterexample to that proposed inference, not an assertion that no useful decomposition is possible with further structure.

5. Event, observation, and parameter obligations

Events and adaptivity. The main domain is every finite history in $\mathcal{E}^*$. Any adaptive strategy whose realized events lie in $\mathcal{E}$ produces one such history, so pathwise quantification includes it. This rectangular domain can overapproximate physically possible combinations. A restricted history-dependent domain needs an explicit closure argument or sufficient history state; it cannot exclude failures retrospectively.

Admitted input. The object that enters the transition is admitted context or task input, not necessarily the raw submission at a chat box or API. An external admission process may validate, normalize, reject, tokenize, or limit a submission. The paper imposes neither an identity admission map nor a requirement that every submitted token affect computation. The non-degeneration check requires some actual task dependence on admitted input, not dependence on every raw request or irrelevant continuation.

Decoding. Events distinguish admitted task input from the exogenous randomness or decoder controls required to make the complete transition deterministic. Sampled tokens are produced inside that transition; when they re-enter computation, that feedback remains inside its scope. The same event history must drive actual and reference computations in the trace theorem. Equality of possible paths is not equality of their probabilities. Equality of stochastic laws requires a common event-generation law or an independently justified probability-preserving coupling. Identical seed names do not by themselves establish equivalent randomness semantics.

For responsiveness, changing the random realization with admitted input held fixed proves nothing. Even comparing different inputs at a fixed random seed can be misleading: a fair bit ω and output $u \oplus \omega$ give different seed-coupled outputs for $u = 0, 1$ but the same output distribution. The revised condition compares actual trace laws under prescribed admitted inputs and a fixed decoder policy and randomness law. The deterministic witnesses have distinct point-mass laws and satisfy it directly.

Endogenous privileged tokens. Fixing p at initialization excludes repeated external privileged intervention. It does not forbid the model from generating, regenerating, or re-entering privileged tokens to sustain the initialized regime. Such endogenous self-stabilization is permitted and its full recurrent effect is part of F_Θ . Internal origin alone does not certify authority, and public token spelling alone does not determine its dynamical effect. The theorem applies only if the complete transition, including that re-entry, satisfies the declared conditions. It neither prescribes a token mechanism nor establishes that a model can sustain the conditions indefinitely.

The pathwise bound must hold for every declared decoder realization, including those that omit any particular proposed stabilizing token sequence. High-probability success does not imply this every-realization guarantee. Full-support decoding may therefore enlarge the obligations substantially, including feedback through self-generated tokens. It does not by itself prove that every nontrivial geometric authority condition is impossible. Nor does pathwise latent persistence imply a universal restriction on emitted tokens unless the operational readout condition establishes that restriction. The numerical witness is deterministic and supplies no stochastic language-model attainability result.

Observations. Y_Θ must include every output for which preservation is claimed. A hidden bypass, omitted tool decision, or unobserved output segment is outside the demonstrated relation. Complete-state fidelity and observation coverage are modeling obligations; an algebraic proof cannot verify that an abstraction matches an unexamined deployed system.

Interface and integrity scope. The model boundary may have finite token or context limits. Quantification over all finite event histories does not require unlimited simultaneous context or admission of unbounded input magnitude. Conversely, a finite context limit alone does not establish a persistent trajectory bound: admitted influence and internal feedback can accumulate across steps. Fixed parameters and a protected initialization exclude adversarial changes to weights or infrastructure through channels outside the declared interface. These are domain and integrity assumptions, not the self-stabilization proof. Finite context capacity alone does not establish that initialization information is eventually erased: that conclusion needs assumptions about window management and every other retained part of the complete state. If different initializations actually lead to identical complete states and common subsequent events, their subsequent computations coincide; the finite-capacity premise alone does not establish this state equality.

Parameters. a_p, λ_p, d_p , the geometry, reference, and domain are fixed for the stated comparison. The theorem is uniform over histories and time after those choices. A general symbolic theorem needs no universal numerical value. A candidate claim must establish its own constants and dependencies. Rescaling the metric and radius together does not manufacture slack: v, d, a scale together while λ is unchanged. Choosing a different observable or reference changes the claim itself.

6. Check of the optional capability comparison

Appendix B of the paper contains the complete optional lemma and proof. No premise or proof needed to assess the manuscript is supplied only by this audit. The comparison retains the main exact authority contract and separately declares a capability baseline B^0, H^0 , task representation Q , task metric ρ , and output metric Δ .

With uniformly bounded one-step task mismatch η , baseline task Lipschitz constant L , one-step readout mismatch ν , and baseline readout Lipschitz constant K , define

$$\varepsilon_t = \rho(Q(\Sigma_t), x_t^0).$$

The triangle inequality gives $\varepsilon_{t+1} \leq \eta + L\varepsilon_t$. Induction and the readout bounds give

$$\varepsilon_t \leq L^t \varepsilon_0 + \eta \sum_{j=0}^{t-1} L^j, \quad \Delta(y_{t+1}, y_{t+1}^0) \leq K\varepsilon_t + \nu.$$

The indexing is consistent: outputs at $t+1$ use the respective states at t and the common event e_t . The initial error is not silently set to zero. In the release candidate, the candidate mismatch bounds hold uniformly on all R_p and events, while B^0, H^0 are total and their Lipschitz bounds are global on $\mathcal{X}^0$, uniformly over events. Theorem 6.1 supplies actual-state containment, and the baseline bounds require no additional trajectory-domain hypothesis. This is a strengthening of the optional lemma's hypotheses; the previous domain-qualified lemma was valid but left domain coverage to be established separately.

For $L < 1$, the task error has a uniform upper bound. For $L = 1$, the accumulated term is $t\eta$. For $L > 1$, it may grow geometrically. A finite-horizon estimate is not an indefinite preservation guarantee. A small hidden-state distance does not establish linguistic coherence, and a discontinuous decoder can invalidate the readout Lipschitz hypothesis. With the discrete output metric, a strict bound below 1 entails exact observation equality; larger bounds may be uninformative.

No universal quality score or acceptable tolerance follows. The lemma expresses a declared comparison if its assumptions hold; it does not prove a useful practical tradeoff exists. If one instead replaces the main identities for the same B_p, H_p with approximate identities, one changes the authority contract. Geometry alone then does not preserve exact initialized-operation authority.

7. Disposition of the adversarial reviews

7.1. Review of v1.3.0

Finding	Audit disposition
F1: the global loss hypothesis carries the difficult trajectory claim	Accepted. Version 1.4.0 derives a history bound from a one-transition inequality. The report's stronger assertion of necessarily identical verification cost is unproved. No complexity advantage is claimed here.
F2: a margin can be irrelevant to privileged task influence	Accepted as the principal theoretical gap. The old prose rejected dishonest interpretations but did not adequately formalize the bridge. The new actual-transition and readout identities provide that bridge for specified operations.
F3: the old positivity requirement is a safety property	Accepted. Reachability from an initial state differs from invariance of every state in a larger region. The new proof uses sufficient transition conditions. A “one-bit non-interference” equivalence needs declared observations, executions, and comparison semantics.
F4: title-level terminology exceeds the mathematics	Accepted for the old interpretation. The new title and claim ledger state the narrower operation-authority result. No general root-trust or prompt-authority theorem is inferred.
F5: decoder choices enlarge the universal domain	Accepted. The further claim that nontrivial satisfaction requires decoder truncation is not proved.
F6: public copies and interface scope need consistency	Accepted as a specification clarification. Restricting an interface to its declared input type is not automatically secrecy; the witness explicitly admits both public root values as later inputs.
F7: domain, terminal, fidelity, and boundary details	Joint events, total reference maps, and declared terminal behavior resolve the relevant formal ambiguities. A zero level set need not have measure zero; no measure or regularity assumption justified that report claim. Blanket claims that every vanishing margin lacks finite-precision meaning are also unsupported.
F8: the audit was not supplied	The paper must carry its own assumptions and proofs. The audit is optional supplementary verification history. Its absence did not invalidate the independent report's substantive findings.

The report's alternating-margin example correctly shows that the old fixed-reference certificate was sufficient rather than necessary: uniformly positive actual margins can fail its stronger reference-relative bound. The $L=0$ and classifier examples expose weak interpretation, not algebraic errors. A correct conditional theorem cannot have a counterexample satisfying its hypotheses; this fact does not make all mathematical implications meaningless. The questions are whether the assumptions have independent content and whether their conjunction has relevant instances.

7.2. Checking the v1.3.0 proposed repairs

R1 requires a base case. Its condition preserves $m \geq a$ but assumes only that initialization is in a larger domain. Set $D = \mathbb{R}$, $m(x) = x$, $a = 1$, $x_0 = 1/2$, and $F(x) = x$ for $x \geq 1$, $F(x) = -1$ otherwise. The proposed invariant-region condition holds, but the initial run immediately loses positivity. Add $m(x_0) \geq a$. The new paper supplies its corresponding initialization condition explicitly. A multiplicative decay variant also needs its factor and closure assumptions stated.

R2 is useful but insufficient as proposed. An independent override relation and a sound certificate can supply meaning without introducing task correctness. Merely requiring some output dependence on p permits an irrelevant header. Also, an honest override classifier can satisfy the proposed soundness condition, so R2 does not exclude every W-D classifier construction as claimed. The new paper connects authority to the identified actual computation instead of relying on generic causal sensitivity alone.

R3 does not establish dominance. Suppose authorized, competing, and unauthorized continuations are distinct deterministic outputs A, B, and C. Total-variation distance from B is 1 for both A and C. Remaining far from a competitor therefore does not establish preservation of the authorized operation. A nuisance watermark supplies another distinguishability witness with no authority over substantive content.

The report's claim that making privilege independently meaningful necessarily abandons a geometric thesis in favor of a behavioral or counterfactual thesis is not established. A geometric condition can have a specified operational interpretation without an answer-quality criterion. The valid demand is an independent meaning for the protected relation. The present repair supplies one, and openly limits its scope.

7.3. Review of v1.4.0

The subsequent review accepted the core proofs and the non-vacuity of the abstract construction. It raised interpretive and scope objections rather than an algebraic counterexample satisfying the stated hypotheses. Version 1.4.1 retains the core implications and revises the points below.

Finding	Audit disposition
F1: exact task matching induces congruence along equal- q states	Accepted. Equal represented task states in the same privileged region have equal next represented states and equal observations for a common event. The further claim that geometry must always lie outside task state or be behaviorally inert does not follow. Proposition 7.4 gives a nontrivial $q = g = \text{id}$ counterexample.
F2: preserving a reference can preserve behavior described informally as an override	Accepted as a substantive scope limit. The result preserves a specified operation; it does not interpret arbitrary prompt authority. A reference retrospectively defined as the candidate itself violates the independent-specification requirement and is not evidence against that requirement's stated form. Neither point establishes transformer attainability.
F3: independence of the reference requires an external specification	Accepted as an obligation on the comparison. Equal function values alone cannot reveal whether the reference was selected retrospectively. The stronger assertion that independent specification is universally impossible to formalize is unsupported.
F4: the unrestored comparator is weak	Accepted as a limitation of relevance. It remains a valid declared comparator for a strict, limited mathematical existence result. The paper identifies it as an abstract comparison and makes no claim about competitive language models.

Finding	Audit disposition
F5: exact preservation is assumed locally and no practical tradeoff is proved	Accepted. Local matching alone does not establish joint satisfiability with geometric persistence; Proposition 7.1 supplies that missing existence fact. Exact reference preservation is a sufficient zero-loss case, not a theorem that robustness has no capability cost or that useful shared-resource realizations exist.
F6: event variation can mistake decoder noise for input responsiveness	Accepted and repaired. Some variations in admitted task input must alter the actual task-output trace law under a fixed decoder policy and randomness law. Variation only in decoder randomness is insufficient. Requiring different transition formulas at a common state is not necessary when initialization and a fixed interpreter realize the declared task distinction.
F7: the strict budget is stronger than finite-step invariance requires	Accepted as a sufficiency limitation, already visible in the boundary example. The strict condition supplies a uniform positive margin. At equality, some systems stay inside at every finite step without a uniform margin; others can hit the boundary. No necessity claim is made.
F8: the audit was not provided with the paper	A standalone paper is the appropriate review target. Appendix B now supplies the optional capability-comparison lemma in full. This audit records checks and revision history; it is not required to fill gaps in a core proof.

The decoder-only witness is a real regression. If events are (u, ω) , and a task emits $p\omega$ while ignoring u , output can vary across events and privileged configurations without any input responsiveness. The old event-history wording allowed that reading. The new trace-law condition excludes it without requiring every admitted input to affect every continuation. It also respects the distinction between provided input and admitted context.

The mathematical and engineering objections remain separate. The abstract construction establishes joint consistency, and the shared-state example removes one claimed structural necessity. Neither establishes useful transformer weights or the mapping from privileged language to the independently specified operation. Persistent endogenous self-stabilization, including privileged-token feedback, is permitted by the formalism; satisfaction of its full-transition bound in a neural model remains unproved. A finite context window and protected infrastructure delimit the interface but do not discharge that obligation.

7.4. Review of v1.4.1 and release-candidate disposition

The latest report again verified the core theorems, both propositions, the numerical instances, and Lemma B.1. It explicitly retracted the previous universal claim that g must be observationally inert. No satisfying counterexample to a stated theorem was supplied. Its useful remaining clarifications do not require a foundational rewrite.

Finding	Release-candidate disposition
F1': factorization, decomposition, confinement, and a proper quotient	Accept the conditional $g = h \circ q$ reduction on $q(R_p)$ and its observable-confinement consequence. H-op and the factorization remain candidate obligations. Reject the unproved general independent decomposition and the claim that proper-quotient shared geometry is open; Section 4.5 supplies direct examples.

Finding	Release-candidate disposition
F2': in-spec triggers and the proposed name “mode locking”	Retain the acknowledged interpretation limit. The protected object is the specified task operation, which can itself include mode changes; “mode locking” would obscure that scope. A retrospectively chosen reference remains inadmissible.
F3': endogenous stabilization under every realization	Accept and state explicitly that the complete bound covers every declared decoder realization, including ones missing particular stabilizing tokens. Reject the inference from bounded context alone to erasure of initialization or a unique carrier of persistence.
F4': either copied tokens stabilize or reserved provenance markers are necessary	Reject the binary. Public copies may oppose the initialized regime yet remain unable to displace it. The existing deterministic witness admits repeated $u = -1$ at $p = +1$ and retains $r_t = 1/2 + 2^{-(t+1)} > 0$, with no provenance variable or reserved marker.
F5': geometric confinement constitutes a capability cost	Reject the inference. A bounded selected observable does not establish task-complexity or comparative-capability loss. Joint satisfiability remains a meaningful abstract result; no practical cost curve or shared-resource language-model result follows.
F6': optional comparison domains	Accept the simplification. Appendix B now requires global baseline Lipschitz bounds and uniform candidate mismatch bounds throughout R_p . The original conditional lemma was correct; its separate domain-coverage obligation is removed by stronger hypotheses.
F7': echo, equal-length histories, and independent specification	Echo is a legitimate simple task, and the minimum responsiveness test expressly does not establish competence. Do not restore an undefined “beyond retention” condition. Equal-length input histories exclude trace-length artifacts. The independent-specification obligation remains explicit.
F8': equality, smaller checking regions, and cross-reference	Retain the sufficient strict budget and stated region. Correct “Section 7.4” to “Proposition 7.4.” These optional weakenings supply no needed repair to the valid implications.

The report's phrase “verifiable on the reference alone” is sound only after the operational relation, factorization, and relevant image have been established. Its further claim that the candidate-specific obligation is empty does not follow. For example, increasing only the candidate's gain from $1/4$ to 1 breaks matching to the fixed reference of Proposition 7.4. Increasing the reference's gain as well defines a different comparison.

These revisions close the identified specification ambiguities while retaining the proof scope. They do not turn the abstract existence results into claims of neural attainability, a useful universal quality metric, or a construction method.

8. Historical register: v1.2.0 mathematical objects

The obsolete objects below remain removed or narrowed. This register preserves the prior audit's numbered findings; it does not propose restoring the old machinery.

Source object	Finding and current disposition
Theorem 3.1	Exact equality to a vector outside the embedding span is impossible at that linear map. This is neither orthogonality nor dynamical unreachability; sequence aggregation needs specification. Removed from the foundation.
Definition 6.1	Linear subspaces and nonlinear manifolds do not share projection algebra. An unchanged manifold need not preserve authority. Replaced by typed geometry and an operational relation.
Definition 6.2	$D^\dagger$ was called an adjoint without linearity or inner products; D also changed type from operator to subspace. Removed from the proof.
Definition 6.3	“Functions or vector fields” did not identify the invariant object. Topology, field values, cones, and basins are different claims. Replaced.
Definition 6.4	“Governed primarily” lacked a comparison. Both admitted-input effects and initialized evolution arise through fixed dynamics. Replaced by explicit hypotheses.
Definition 6.5	Projected total step displacement is not input-induced opposing influence. It does not prove prefix exclusion or dominance. Removed.
Definition 6.6	The kernel tuple needs consistent component types and predicates. Its remark confused the conjunction of fixed parameters and satisfied conditions. Replaced by the declared system data.
Axiom 1	A nonzero derivative at every step is not general responsiveness; it excludes nondifferentiable and discrete dependence and conflicts with autonomy. Replaced by observed task distinctions and preservation.
Axiom 2	Equal Betti numbers do not imply homotopy equivalence. Perturbed images need not be manifolds; curvature and injectivity radius need structure, extrema need not exist, and $\infty - \infty$ is undefined. Removed.
Axiom 3	A global one-step bound permits accumulated drift and can conflict with unbounded linear recurrences. Removed as a dominance premise.
Theorem 7.1	The $d/t - s$ temporal bound follows by telescoping for a linear projection. It implies neither compact containment nor continuous-time smoothness. Narrow fact only.
Optional Axiom 3a	Same-state input Lipschitzness does not bound repeated trajectory sensitivity without state-dynamics assumptions. It does not supply unrestricted “no cliff” behavior. Removed.
Axiom 4	The parameter was unused and $\gg$ lacked direction, constants, and quantified meaning. Total update norm cannot establish relevant influence or persistence. Replaced.
Axiom 5	$\partial R / \partial D$ was undefined with overloaded set/operator symbols. Current readout nonmutation does not cover autoregressive re-entry. Replaced by full transition and observation scope.
Axiom 6	Sole-path mediation is conditional and does not itself attenuate influence. An interface name or diagram proves no dominance bound. No driver architecture is imposed.
Axiom 7	Outside-span and current decoder-null vectors are unnecessary for public shared-space privilege and insufficient for persistence. Removed.

Source object	Finding and current disposition
Theorem 8.1	Duplicates the narrow embedding equality fact of Theorem 3.1; “concepts” are not confined by that argument. Removed from core.
Theorem 8.2	$W_U(h + ar) = W_U h$ is valid when $W_U r = 0$. It covers the stated current linear readout, with matched decoder state, not future dynamics or an omitted nonlinear transform. Removed from core.
Theorem 8.3	$\dim R > \dim U$ does not constrain independent W_U . Output-null directions need not exist. Separate existence claims also need not supply one vector satisfying both conditions. Rejected.
Definition 8.4	Exact autonomy is well-defined for the specified linear direct sum but conflicts with responsive privileged computation. Removed as a requirement.
Theorem 8.5	Autonomous evolution follows by induction. Zero-state unreachability additionally needs $r_0 = 0$ and $f_R(0) = 0$. It is not a dominance result. Removed from core.
Example 8.6	Does not establish all-axiom compliance: Read Access fails; global bounded linear motion generally fails unless $A_{RR} = I$; the unembedding is unspecified. Removed.
Theorem 8.7	The block recurrence establishes its stated mediated dependence, not attenuation, boundedness, or a bidirectional interface. D changes type. Removed architecture example.
Definition 9.1	One constant over the entire admissible domain is a global bound there, not merely local Lipschitzness. The formula concerns full-state dependence at fixed input. Removed from core.
Theorem 9.2	Discrete curvature and the limiting operation were undefined; update norms do not establish curvature or dominance. Rejected.
Theorem 9.3	Bounded increments imply at most td displacement, not one compact region over arbitrary horizons. Other geometric axioms do not repair the proof. Rejected and replaced by Theorem 6.1.

The historical counterexamples remain elementary: $e_1 + e_2$ is outside $\text{span}(e_1)$ but not orthogonal; $Ae_1 = e_2$ transfers influence; $W_U = I$ has no nullspace regardless of a smaller embedding rank; a coordinate swap exposes a previously invisible direction; and $r_t = r_0 + tde_1$ has bounded increments with unbounded displacement. Large motion in an irrelevant coordinate cannot protect a small margin in another. Equal rational Betti numbers, as for a point and the real projective plane, do not determine homotopy type.

Cross-cutting dispositions also remain: use complete running state, include decoder re-entry, distinguish dimensions from topology and reachability, remove universal prefix-erosion claims, and do not equate lexical projection loss with all computational error. Nested rings do not inherit embedding exclusion without conditions. Mediation does not bound amplification. Builder control does not prove weight attainability. Stronger mathematical constraints restrict a realization class but do not prove a monotonic loss of useful capability. Unsupported priority, universal composition, and backwards-compatibility claims remain removed.

9. What would defeat the revised claim?

1. **Implication failure:** a fully specified instance satisfies the local hypotheses and initialization but exits the region or mismatches the reference trace. This would falsify the written theorem or expose an overlooked hypothesis.
2. **Witness failure:** a displayed fixed map fails a local inequality, uses repeated external privileged reinitialization, has no actual admitted-input dependence, or emits something other than its stated task output. For the shared-state example, an overlooked separation between the claimed task and geometric state would also defeat its purpose. Direct algebra checks found none of these defects.
3. **Interpretation failure:** a claimed protected operation, reference, observation, or task domain does not represent the authority and capability being advertised. The fixed-law SWITCH example already defeats an unrestricted natural-language extension.
4. **Realization failure:** a proposed neural model fails the declared conditions or loses substantive capability. That defeats that realization claim; the abstract witness establishes no neural promise to invoke in its defense.

The revision earns an elementary joint existence and preservation result in an explicit mathematical class. It permits nonzero input coupling, nontrivial actual task responses, and a shared task and geometric state. It also gives a restricted comparator showing improved resistance without task loss on a predeclared nontrivial workload. Those facts are meaningful; they do not establish that an arbitrary language model can be made robust while retaining comparable quality.

The research objective remains increased resistance to adversarial displacement while retaining coherence and task-relevant capability comparable to a declared baseline, with capability and generality costs acknowledged as a tradeoff. Systematic incoherence or task collapse would fail that objective. The next theoretical burden is to connect useful privileged conditioning to the specified-operation relation without selecting a self-justifying reference or omitting substantive task behavior. Neural realization and persistent self-stabilization remain separate unresolved burdens.